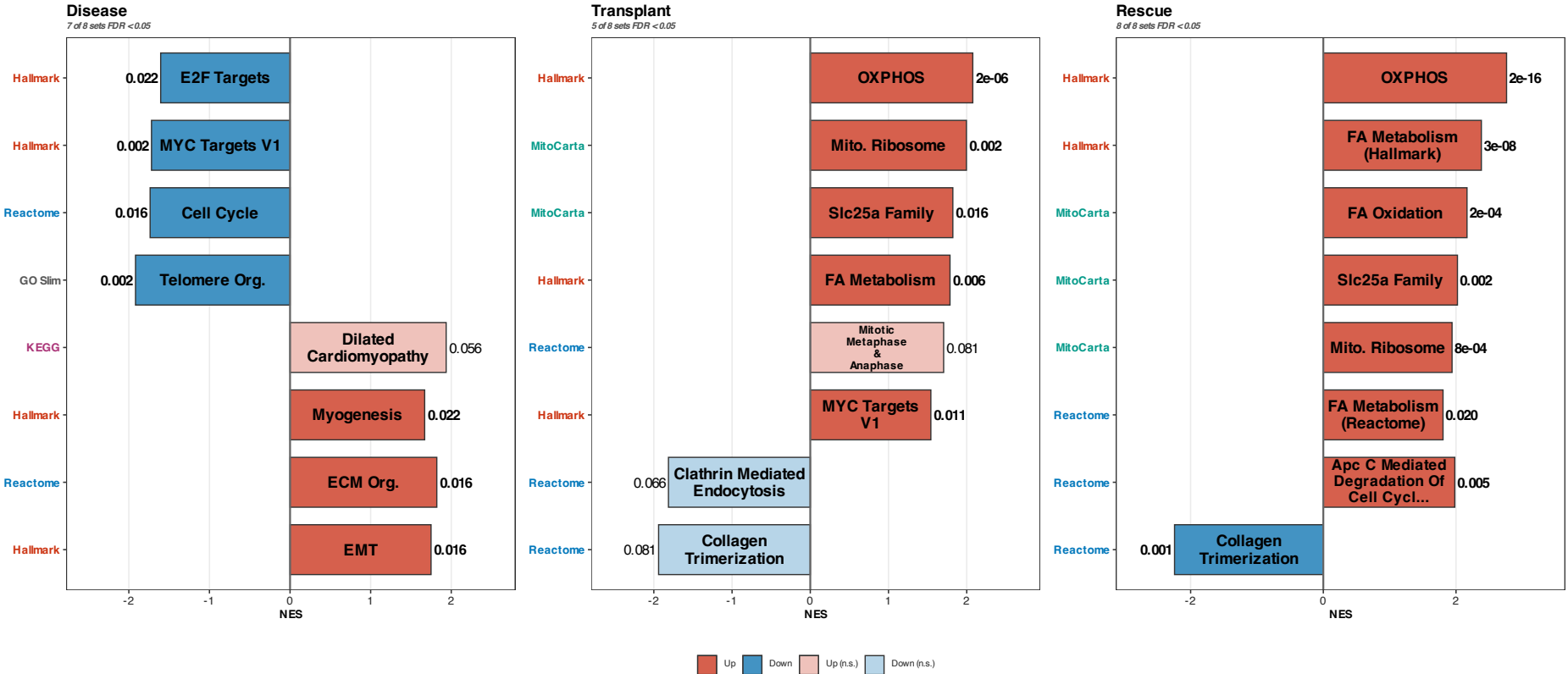


S2 Figure A

Pathway enrichment by contrast

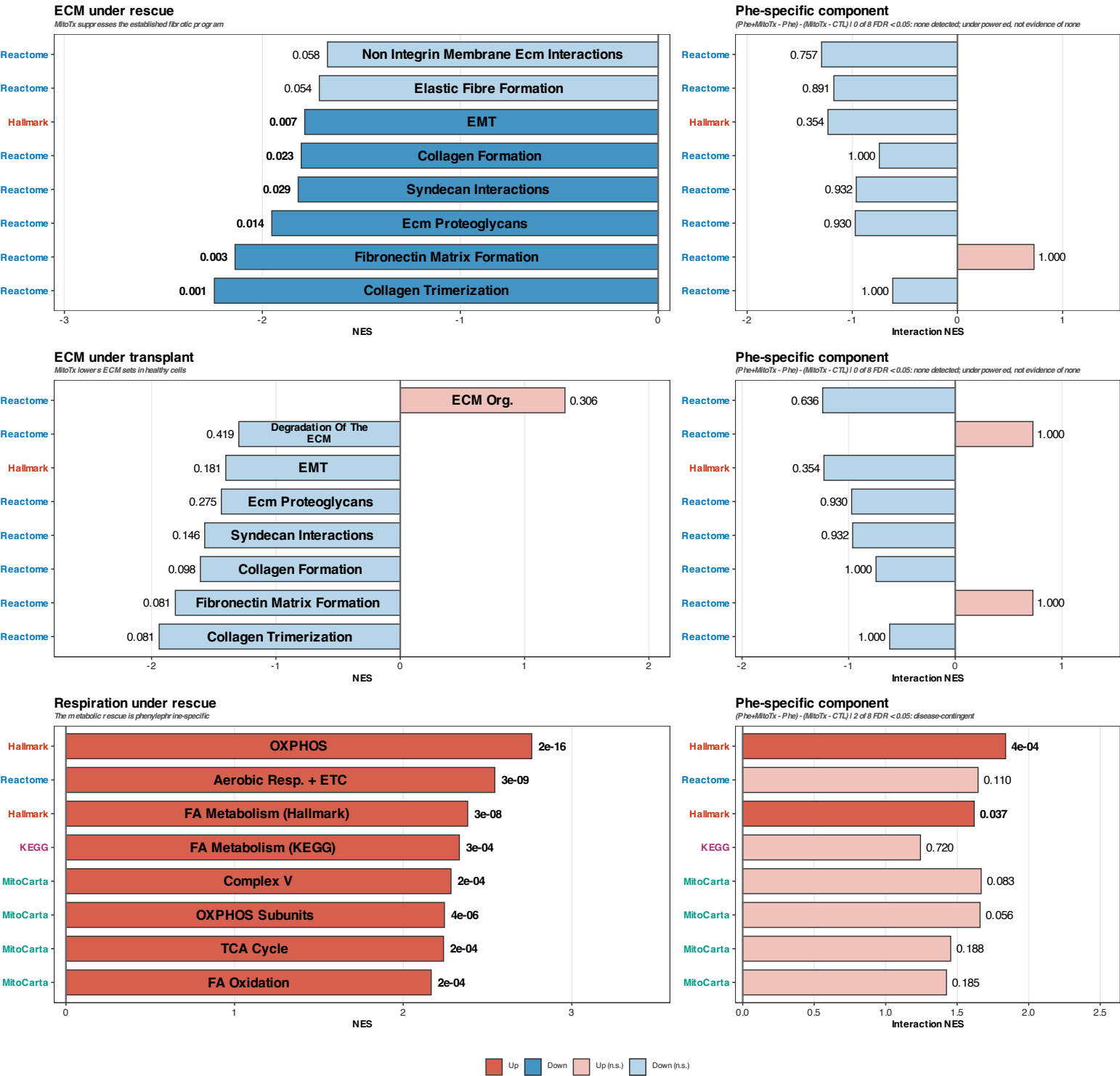
Top 8 themed gene sets per contrast — Disease (Phe vs CTL), Transplant (MitoTx vs CTL), Rescue (Phe+MitoTx vs Phe), fgsea NES, per-database BH-FDR, EnrichmentMap-deduplicated. Sets trace to the fgsea supplementary table.



S2 Figure B

Disease-specificity of the transplant response

Each read shows its main contrast beside the interaction (Phe+MitoTx - Phe) - (MitoTx - CTL), which isolates the disease-contingent component. fgsea NES, per-database BH-FDR; sets from the fgsea supplementary table, not deduplicated so no named set is hidden.



Respiration under rescue

The metabolic rescue is phenylephrine-specific

Set	NES	Direction
OXPHOS	2e-16	Up
Aerobic Resp. + ETC	3e-09	Up
FA Metabolism (Hallmark)	3e-08	Up
FA Metabolism (KEGG)	3e-04	Up
Complex V	2e-04	Up
OXPHOS Subunits	4e-06	Up
TCA Cycle	2e-04	Up
FA Oxidation	2e-04	Up

Phe-specific component

(Phe+MitoTx - Phe) - (MitoTx - CTL) | 2 of 8 FDR < 0.05: disease-contingent

Set	Interaction NES	Direction
Hallmark	4e-04	Up
Reactome	0.110	Up
Hallmark	0.037	Up
KEGG	0.720	Up
MitoCarta	0.083	Up
MitoCarta	0.056	Up
MitoCarta	0.188	Up
MitoCarta	0.185	Up

Legend

Up Down Up (n.s.) Down (n.s.)

S2 Figure C

Disease-selected pathway sets read across contrasts

The four ECM and four proliferation sets phenylephrine moves most under Disease, read under Disease, Rescue and Transplant (names on the Disease column; rows shared). One set clears FDR <0.05 under Rescue; Disease and Rescue share the Phe group and are coupled by design, so this apparent reversal is descriptive only. fgsea NES, per-database BH-FDR.

